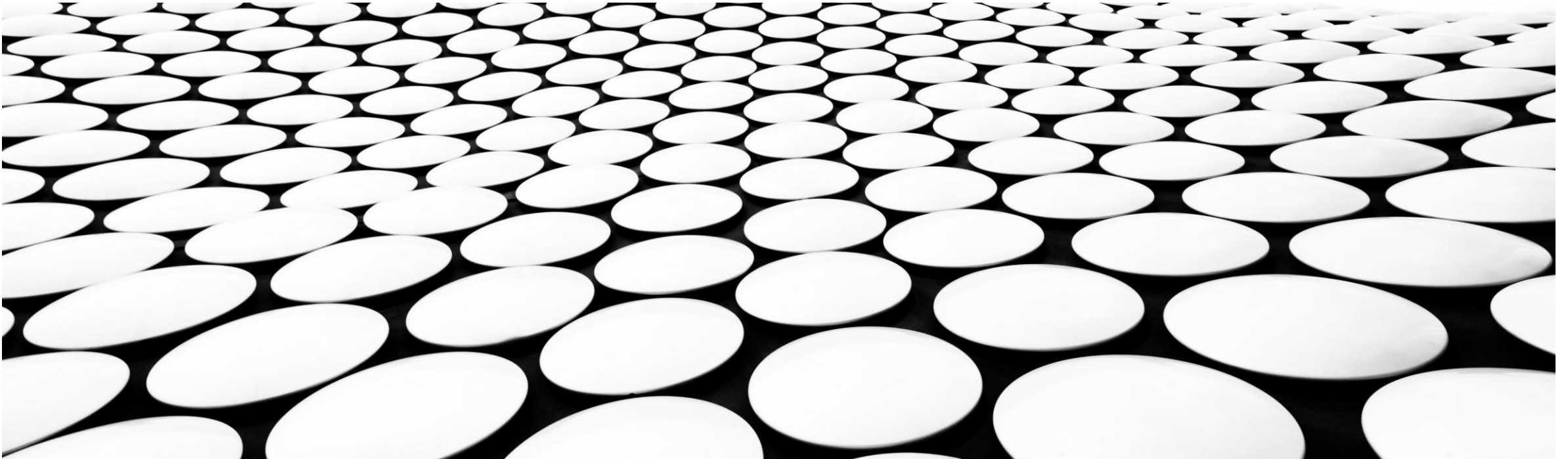
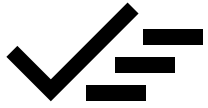


COURSE GOALS AND OVERVIEW

CSI4142 - Fundamentals of Data Science





COURSE GOALS AND OVERVIEW

- Learn to compare multiple sources of information
- Be precise and learn important terms in DS
- Develop your critical skills in DS by asking good questions
- Learn some aspects of DS in depth, and know what you don't know
- Become very knowledgeable of the Data Science Process
- Get hands-on experience using Python programming
- Develop skills valuable in the workplace
- Use Generative AI to help (not hinder) your learning

Explore multiple sources
DEFINITION(S)

Definition 1

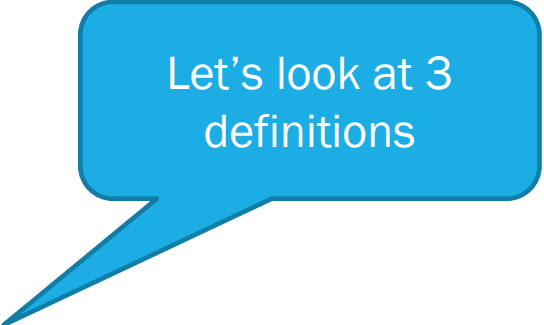
Data science is an interdisciplinary academic field that uses statistics, scientific computing, scientific methods, processing, scientific visualization, algorithms and systems to extract or extrapolate knowledge and insights from potentially noisy, structured, or unstructured data.

Definition 2

Data science combines math and statistics, specialized programming, advanced analytics, artificial intelligence (AI) and machine learning with specific subject matter expertise to uncover actionable insights hidden in an organization's data. These insights can be used to guide decision making and strategic planning.

Definition 3

Data science continues to evolve as one of the most promising and in-demand career paths for skilled professionals. Today, successful data professionals understand they must advance past the traditional skills of analyzing large amounts of data, data mining, and programming skills. To uncover useful intelligence for their organizations, data scientists must master the full spectrum of the data science life cycle and possess a level of flexibility and understanding to maximize returns at each phase of the process.



Let's look at 3 definitions

Definition 1

Data science is an interdisciplinary academic field that uses statistics, scientific computing, scientific methods, processing, scientific visualization, algorithms and systems to extract or extrapolate knowledge and insights from potentially noisy, structured, or unstructured data. *Wikipedia*

Definition 2

Data science combines math and statistics, specialized programming, advanced analytics, artificial intelligence (AI) and machine learning with specific subject matter expertise to uncover actionable insights hidden in an organization's data. These insights can be used to guide decision making and strategic planning. *IBM*

Definition 3

Data science continues to evolve as one of the most promising and in-demand career paths for skilled professionals. Today, successful data professionals understand they must advance past the traditional skills of analyzing large amounts of data, data mining, and programming skills. To uncover useful intelligence for their organizations, data scientists must master the full spectrum of the data science life cycle and possess a level of flexibility and understanding to maximize returns at each phase of the process. *Berkeley*



WIKIPEDIA
The Free Encyclopedia

UC Berkeley School of Information

IBM

Can you link each one to its source?

Know when to stop... is another definition
adding significant information?

What is data science?

Data science is the study of data to extract meaningful insights for business. It is a multidisciplinary approach that combines principles and practices from the fields of mathematics, statistics, artificial intelligence, and computer engineering to analyze large amounts of data. This analysis helps data scientists to ask and answer questions like what happened, why it happened, what will happen, and what can be done with the results.



[What is Cloud Computing?](#) / [Cloud Computing Concepts Hub](#) / [Analytics](#)

What is Data Science?

When searching for information:



Try to search in well-identified and expert sources.



Always try to find the underlying bias... What is objective/subjective?



Stop when new information is redundant information...

Be precise and learn DS terms

GLOSSARY



Data Science Glossary : Definitions for Common Data Science Terms

Get on the path to data literacy with this comprehensive data science glossary: from Activation Function to Z-Score, it's all covered.

Mar 16, 2022 · 38 min read

I like glossaries 😊

A

- Accuracy Score
- Activation Function
- Algorithm
- Apache Spark
- API
- Artificial Intelligence (AI)
- Artificial Neural Networks (ANN)

I

- Imputation

K

- K-Means
- K-Nearest Neighbors (KNN)

L

- Linear Algebra
- Linear Regression
- Logistic Regression

S

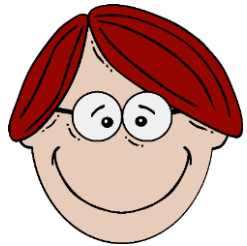
- Sampling Error
- SQL
- Standard Deviation
- Supervised Learning
- SVM
- Synthetic Data

Test your knowledge before and after the course (we can't cover everything... but still...)

data science glossary

algorithm	AngularJS	artificial intelligence	backpropagation
Bayes' Theorem	Bayesian network	bias	Big Data
binomial distribution	chi-square test	classification	clustering
coefficient	computational linguistics	confidence interval	continuous variable
correlation	covariance	cross-validation	D3
data engineer	data mining	data science	data structure
data wrangling	decision trees	deep learning	dependent variable
dimension reduction	discrete variable	econometrics	feature
feature engineering	GATE	gradient boosting	gradient descent
histogram	independent variable	JavaScript	k-means clustering
k-nearest neighbors	latent variable	lift	linear algebra
linear regression	logarithm	logistic regression	machine learning

Before CSI4142!!



We did some analysis to show different types of customers.

Who looks more knowledgeable?



Our data segmentation approach, based on the k-means clustering algorithm, showed 3 different types of customers.

After CSI4142!!

Develop your critical thinking
TRUST



What do you think of this word cloud about Data Analysis?

Trust in Data Collection

Who gathered this data?
What purpose was it gathered for?
What is the sampling method?

As future Data Scientists... be knowledgeable to evaluate the work of others, and to be trusted!

Trust in Analysis method/tools

What are the analytical methods used?
Were reliable tools used to perform the analysis?

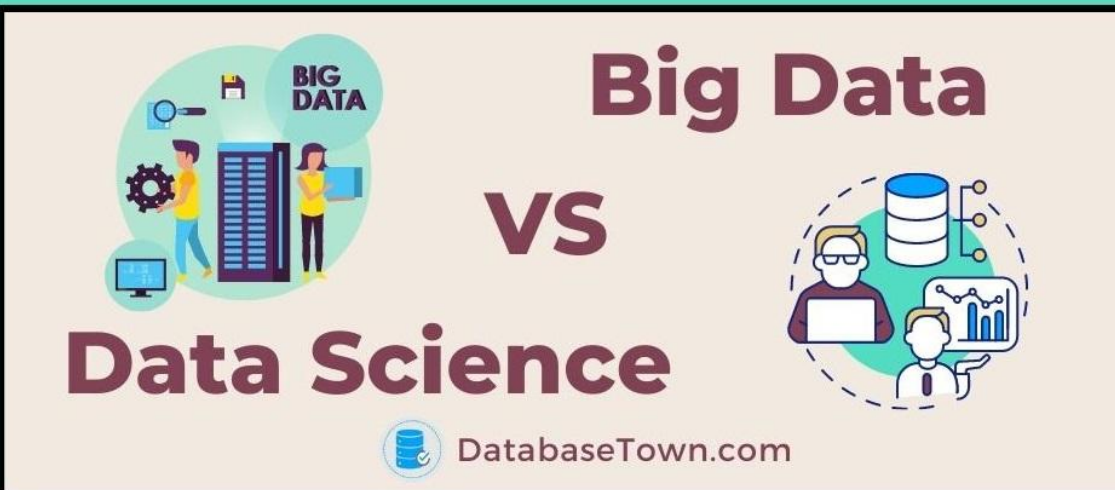
Trust in the Analyst (user of the tool)

Who is using the methods?
What is the expertise of the tool user?

Trust in the result publishing

Who is publishing the results?
Are they telling the whole story?

Learn a bit in depth / Know what you don't know
IN-SCOPE / OUT-OF-SCOPE



Big Data VS Data Science

DatabaseTown.com

Criteria	Big Data	Data Science
Focus	Handling large data volumes and varieties	Advanced analytical modeling and predictions
Emphasis	Data engineering - storage, movement, processing	Data analysis - statistics, machine learning

Even if Big Data is important, this course will not focus on Big Data analysis.



Data Analytics

['dā-tā-nā-'lī-tiks]

The science of analyzing raw data to make conclusions about that information.

Investopedia

Types of Data Analysis



Descriptive Analysis



Inferential Analysis



Predictive Analysis



Prescriptive Analysis



Text Analysis

QuestionPro



Education Nest

Big Data Management Strategies for Efficient Data Storage and Retrieval

Even if Data Engineering is important, this course will focus more on Data Analysis

Data Scientists vs Data Engineers



Zuhailnasir · Follow
8 min read · Jan 13, 2023

Consult other sources to see if they define the roles this way?

Look at job offers on LinkedIn to see if the descriptions are in line with the descriptions found here.

Data scientist: Use various techniques in statistics and machine learning to process and analyse data. Often responsible for building models to probe what can be learned from some data source, though often at a prototype rather than production level.

Data engineer: Develops a robust and scalable set of data processing tools/platforms. Must be comfortable with SQL/NoSQL database wrangling and building/maintaining ETL pipelines.



Data types

In this class, most of our focus is on Tabular Data

Tabular Data

homogeneous
(contains a single type of data)

		columns		
rows		0	1	2
	0	1.5	21.0	76.4
	1	4.0	35.0	99.7
	2	3.0	17.0	85.3
	3	4.0	53.0	90.7

All numerical data

		columns		
rows		0	1	2
	0	'a'	'Mia'	'1'
	1	'c'	'Lucas'	'x-1'
	2	'e'	'Ang'	'zz'
	3	'b'	'Jia'	'0.3'

All string data

heterogeneous
(contains multiple types of data)

		columns		
rows		0	1	2
	0	1.5	21.0	'Mia'
	1	4.0	35.0	'Lucas'
	2	3.0	17.0	'Ang'
	3	4.0	53.0	'Jia'

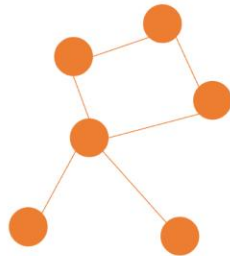
Numerical and string datatypes

Non-tabular Data Examples

Unstructured Text

'Twas brillig, and
the slithy toves
Did gyre and gimble
in the wabe:
All mimsy were the
borogoves,
And the mome raths
outgrabe.

Network Data



Geospatial Data



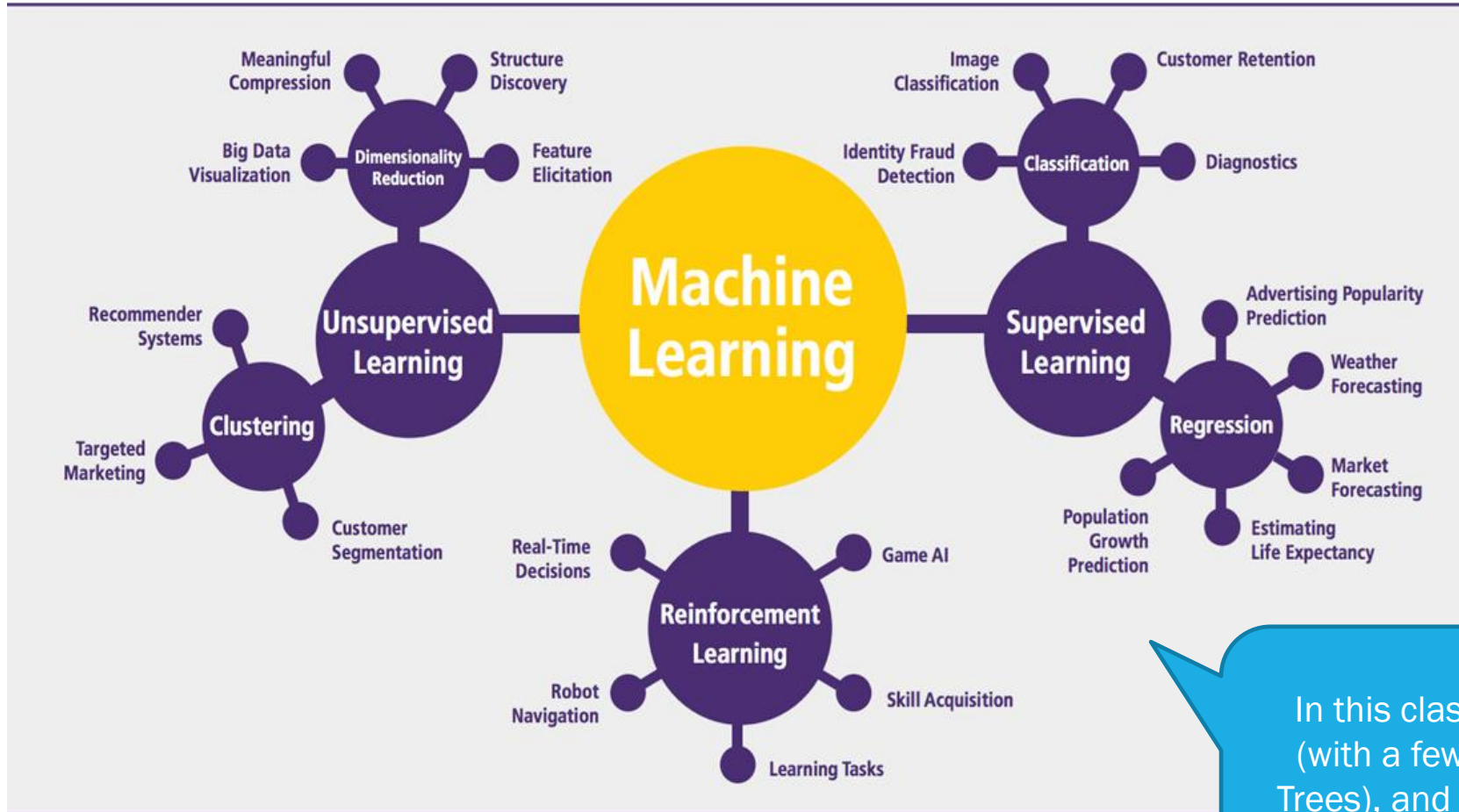
Image Data



Video Data



Supervised ML / Unsupervised ML



In this class, we look at Supervised Learning (with a few specific algorithms, e.g. Decision Trees), and Unsupervised Learning, also with a few specific algorithms (e.g. k-means clustering)

Become very knowledgeable of the
Data Science Process

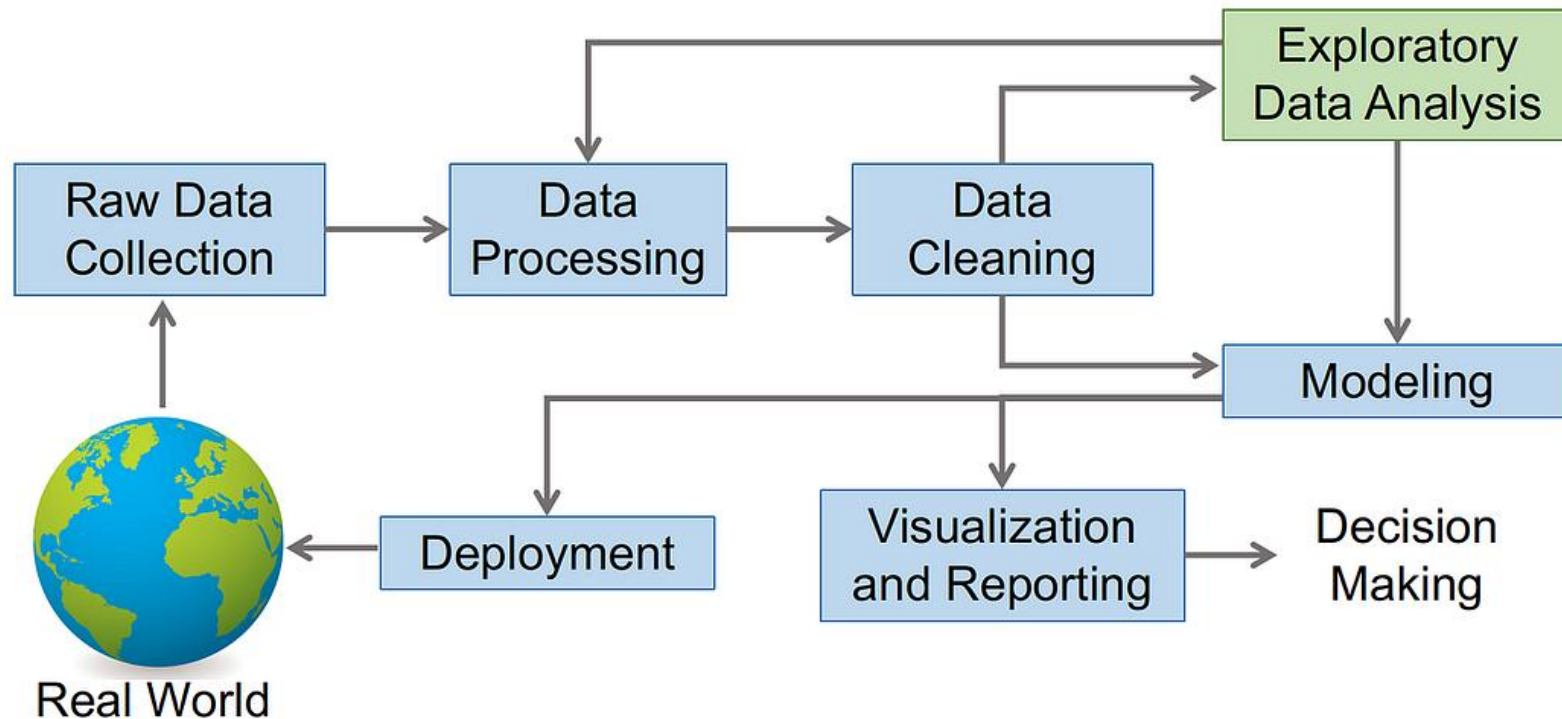
Data Science Process: A Comprehensive Guide



Abhijit · Follow
6 min read · Jan 15, 2024

We'll go through the different steps during the semester, but not with equal emphasis.

Data Science Process



5 Things I Wish I Knew Before My First Job as a Data Analyst



Margaret Efron · Follow

Published in Learning Data · 5 min read · Aug 23, 2023

Many authors who have industry experience, highlight this emphasis on time spent on data cleaning. We will look into that step for sure!

2. You'll spend a lot of time cleaning data.

In data analytics courses, you (mostly) analyze datasets that are already nice and clean.

As a Data Analyst, you analyze messy datasets — surveys from years ago where fields were not formatted properly, values are not correctly capitalized, and the Excel formulas result in errors. You may be asked to create data models for visualizations when none of your coworkers know the source of the data or how accurate it is.

5 Things I Wish I Knew Before My First Job as a Data Analyst



Margaret Efron · Follow

Published in Learning Data · 5 min read · Aug 23, 2023

3. People won't always be happy with your numbers.

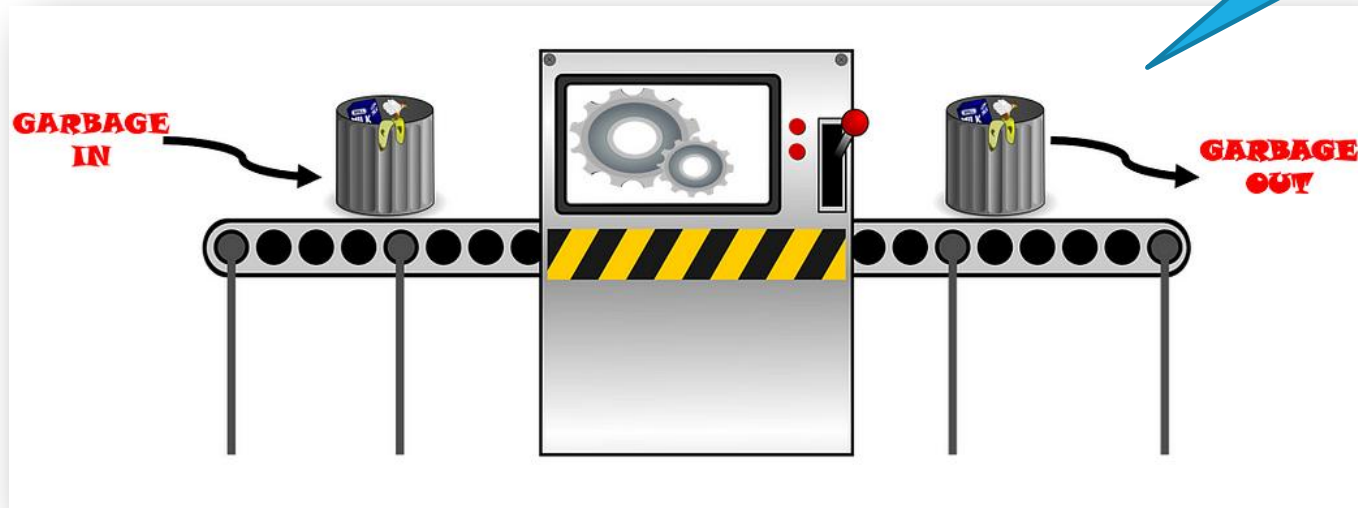
Everyone in business says they want “data-driven insights.” What this means is that they want data points that support their arguments, make them look good, or help them sell their product.

Sometimes your coworkers or clients will not be happy with your numbers. They may ask you to calculate the average customer satisfaction rate, and then be upset when the number is not higher. It does not matter if you tell them to not shoot the messenger — they may still be disappointed, or search for any other metric to report on that will paint their product in a favorable light.

Data analysis SHOULD be data-driven! We'll make sure we understand EDA and statistics and can justify our choice of analysis.



We'll make sure to understand data quality and good data collection methods



Get hands on experience
PRACTICAL WORK WITH PYTHON

DATA SCIENCE AND BUSINESS ANALYTICS

Top Programming Languages For Data Scientists [202]



✕ in 47 Min Read

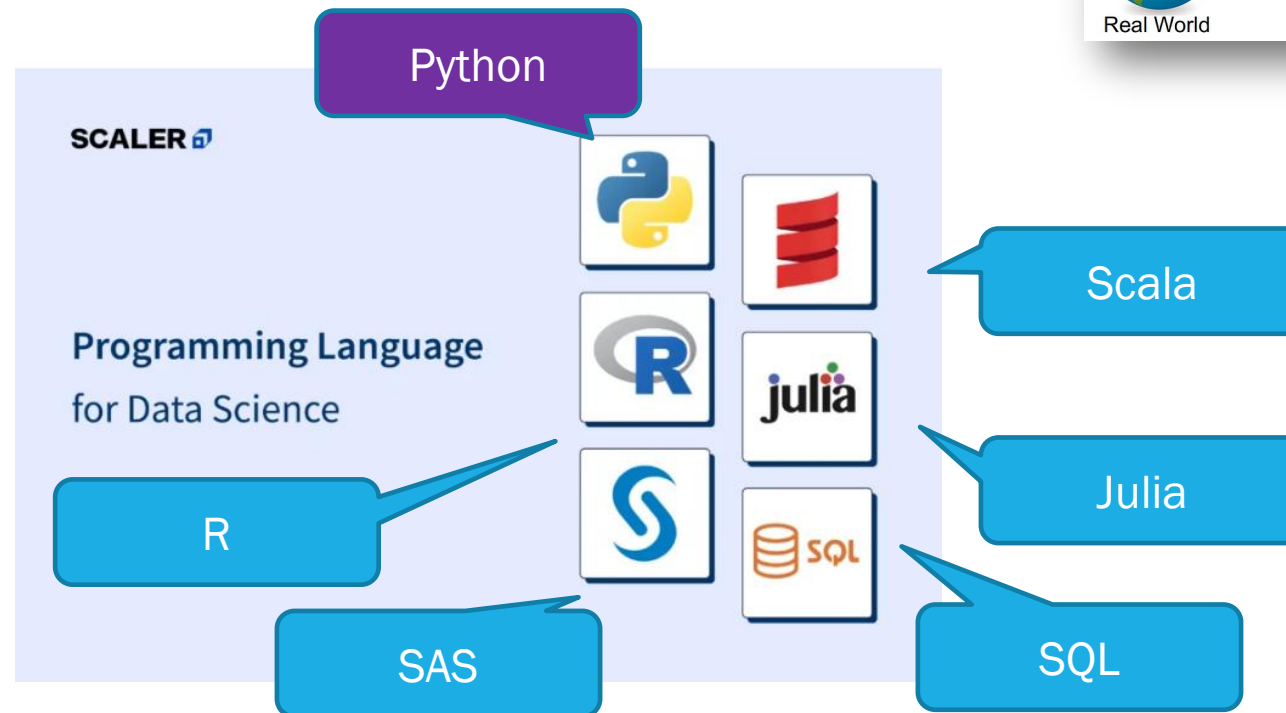
Written by:

SUSHANT GAURAV

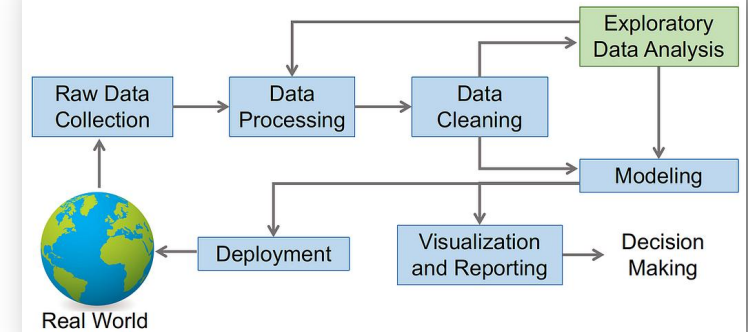
- Senior Analyst @PwC | Technical Content Writer | Ex-Microsoft Learn Student Ambassador

Reviewed by: ANSHUMAN SINGH

Last updated: December 18, 2024 7:52 pm



Data Science Process



Fulfill the *workplace* expectations for your
DATA SCIENCE SKILLS



To begin with Data Science Modelling, the Data Science Companies expect the ideal candidate to know some skills. Below, are the skills one should know before carrying out Data Science Modelling:

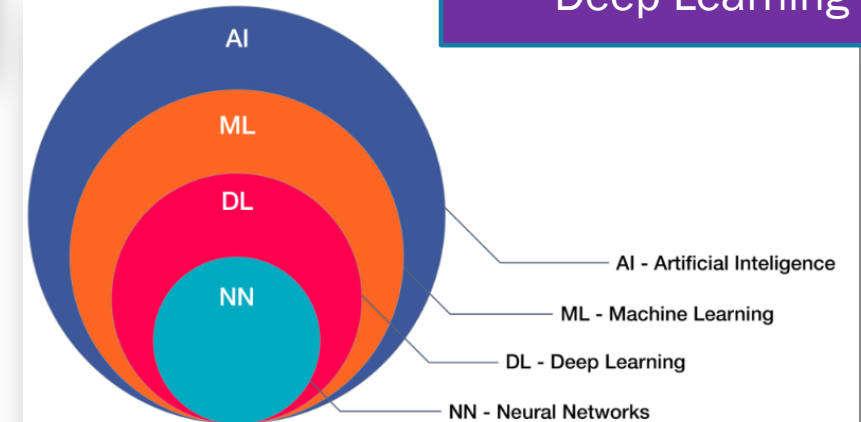
Statistics and Probability



Data Visualization Skills



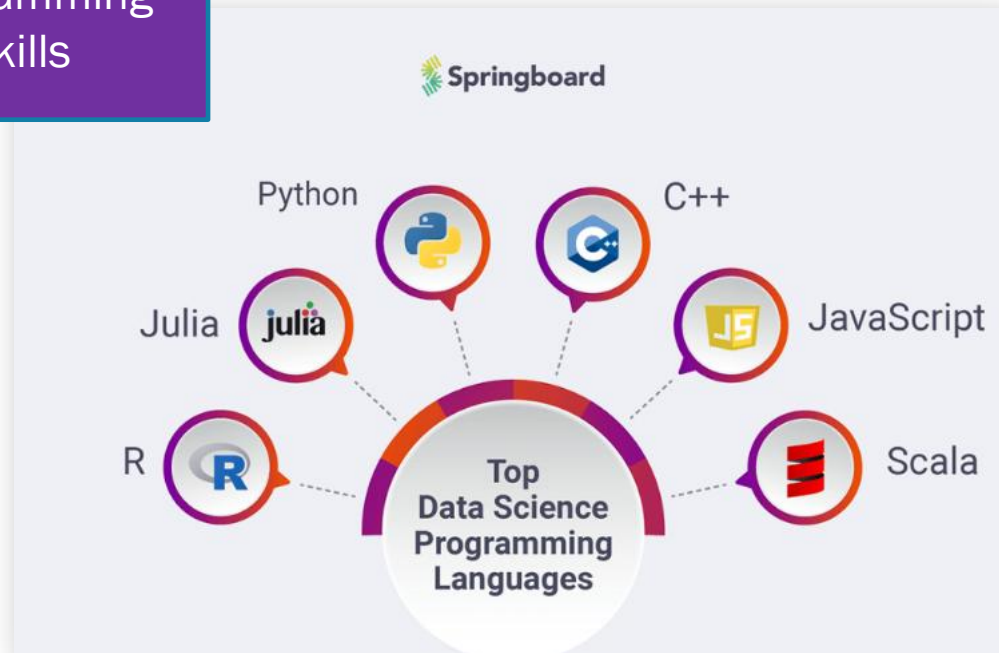
Machine Learning and Deep Learning





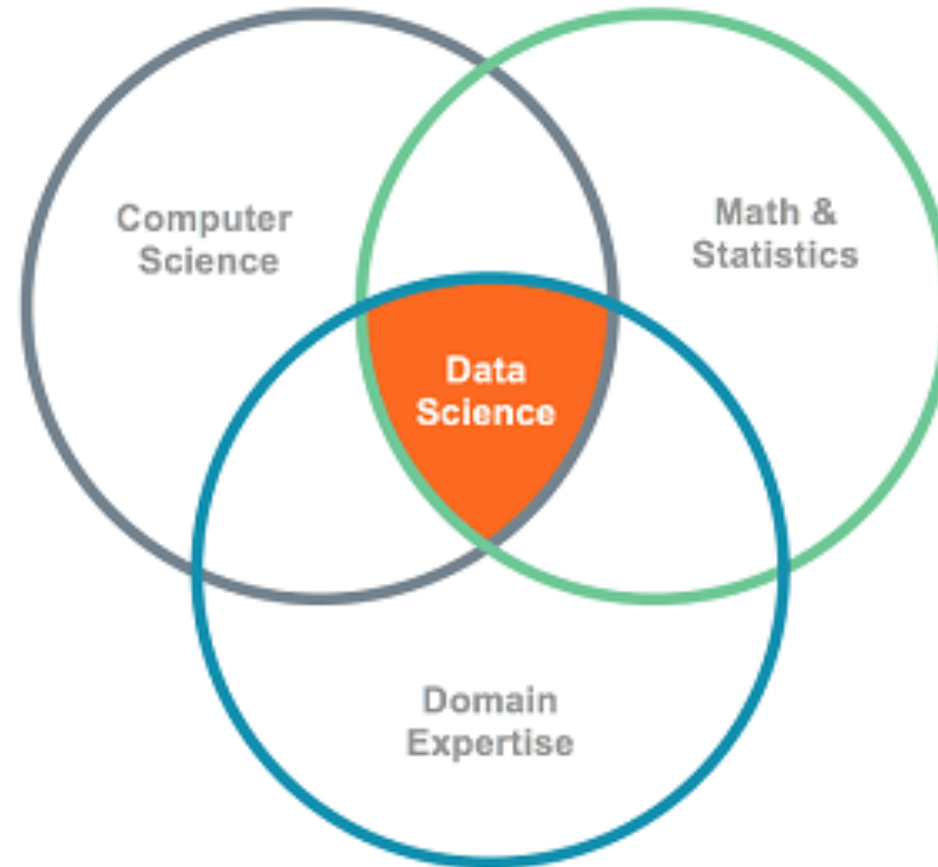
To begin with Data Science Modelling, the Data Science Companies expect the ideal candidate to know some skills. Below, are the skills one should know before carrying out Data Science Modelling:

Programming Skills



Communication Skills





What is shown here that was not mentioned in the skills?

Why Technical Skills (SQL, Python...) Aren't Enough for Data Analysts anymore.



Ethan Duong · Follow

4 min read · Nov 16, 2024

We can't become domain experts in all domains... but we'll explore a few domains during assignments, and as case studies shown in class.

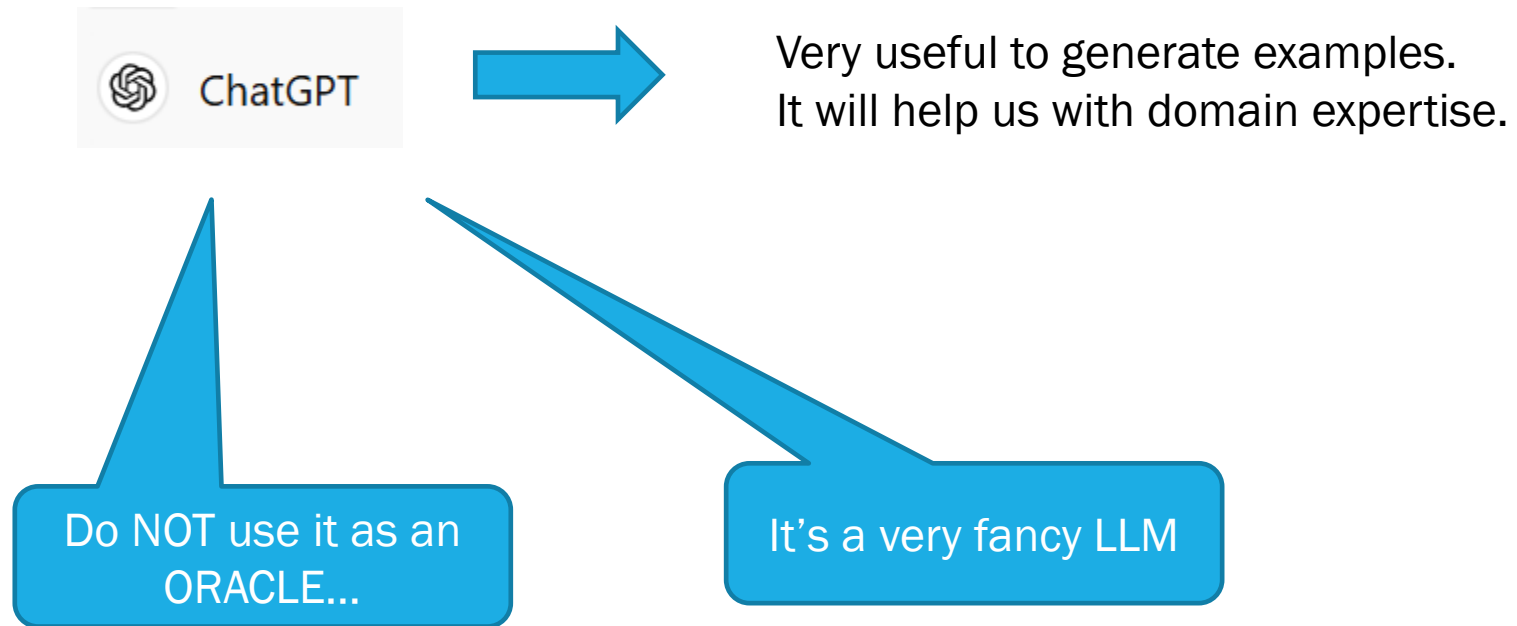
But here's the thing — due to the abundance of accessible learning materials, almost everyone seems to have these skills now. That makes the job market much more competitive.

So, if you want to stand out as a data analyst, you need more than just technical know-how. This is where domain knowledge comes in — it can really set you apart.

Let's take a quick example. Imagine a financial recruiter going through resumes.

- **Candidate A says**, "Excellent at analyzing data to bring insights to business,"
- **Candidate B says**, "Interested in analyzing financial data to bring new insights."

Use Generative AI to help you learn
CHATGPT (or other)



Can you provide 10 case studies for data science, in different domains, in the form of questions.



ChatGPT

1. Healthcare:

How can we predict the likelihood of hospital readmissions within 30 days based on patient demographics, medical history, and treatment details?

2. Environmental Science:

Can we identify regions at high risk of wildfires by analyzing historical fire incidents, weather data, and vegetation patterns?

3. Retail:

Which factors influence customers to abandon their online shopping carts, and how can this be predicted and reduced?

Can you provide 10 case studies for data science, in different domains, in the form of questions.



ChatGPT

4. Finance:

Can we detect fraudulent credit card transactions using customer spending patterns and transaction metadata?

5. Transportation:

How can we optimize the scheduling of public transit to minimize delays and improve service efficiency during peak hours?

Can you provide 10 case studies for data science, in different domains, in the form of questions.



ChatGPT

6. **Education:**

What patterns in student engagement (e.g., attendance, assignment submissions) predict academic success or dropout risk?

7. **Social Media:**

Can we classify and predict the virality of social media posts based on text content, hashtags, and user engagement metrics?

Can you provide 10 case studies for data science, in different domains, in the form of questions.



ChatGPT

8. Sports:

How can we forecast player performance in an upcoming season using historical statistics, physical metrics, and game logs?

9. Real Estate:

Which factors are most influential in determining property prices, and how can we accurately predict prices for a given location?

10. Energy:

How can we forecast daily electricity demand in urban areas using weather data, historical usage patterns, and population density?



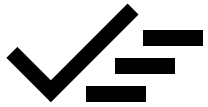
ChatGPT

Can you provide 10 case studies for data science, in different domains, in the form of questions.

Prompt Engineering – How to write good prompts

As « generation is free », we are tempted to ask for too many at a time.

Amazing ressource if we use it wisely.



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